40EWR-L digimatic schnittstelle pruefung - korrektur

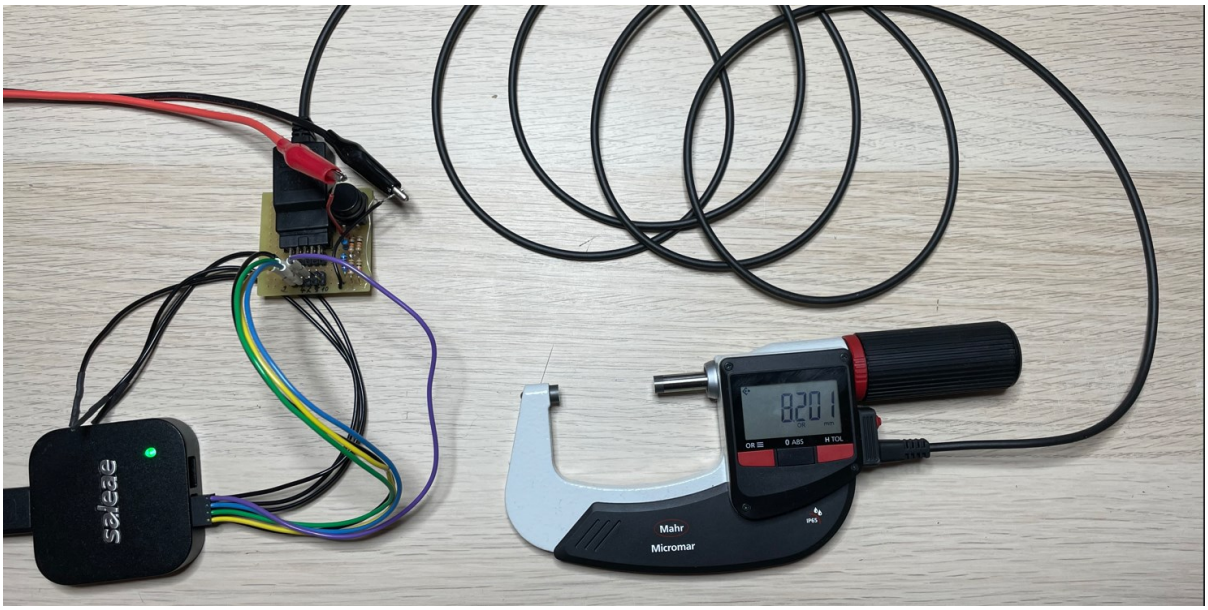
1. Messaufbau:

1.1. 40EWR-L, 25-50mm (art.: 4157021, sn.: 22100002)

1.2. Digimatic Kabel: DK-D1

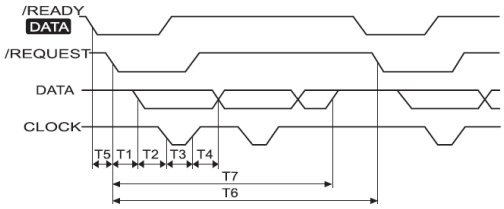
1.3. Messung/Empfänger: Saleae logic Pro 8

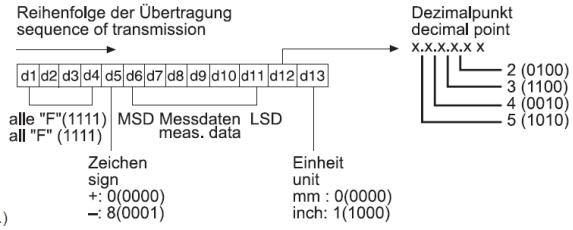
1.4. Signalkonditionierung: 3VDC an DATA, CLOCK und REQUEST



2. Interface Beschreibung

(Datenblatt: Ba_3723295_DK-U-D_de_en_fr_es_it_zh_0322-1.pdf):

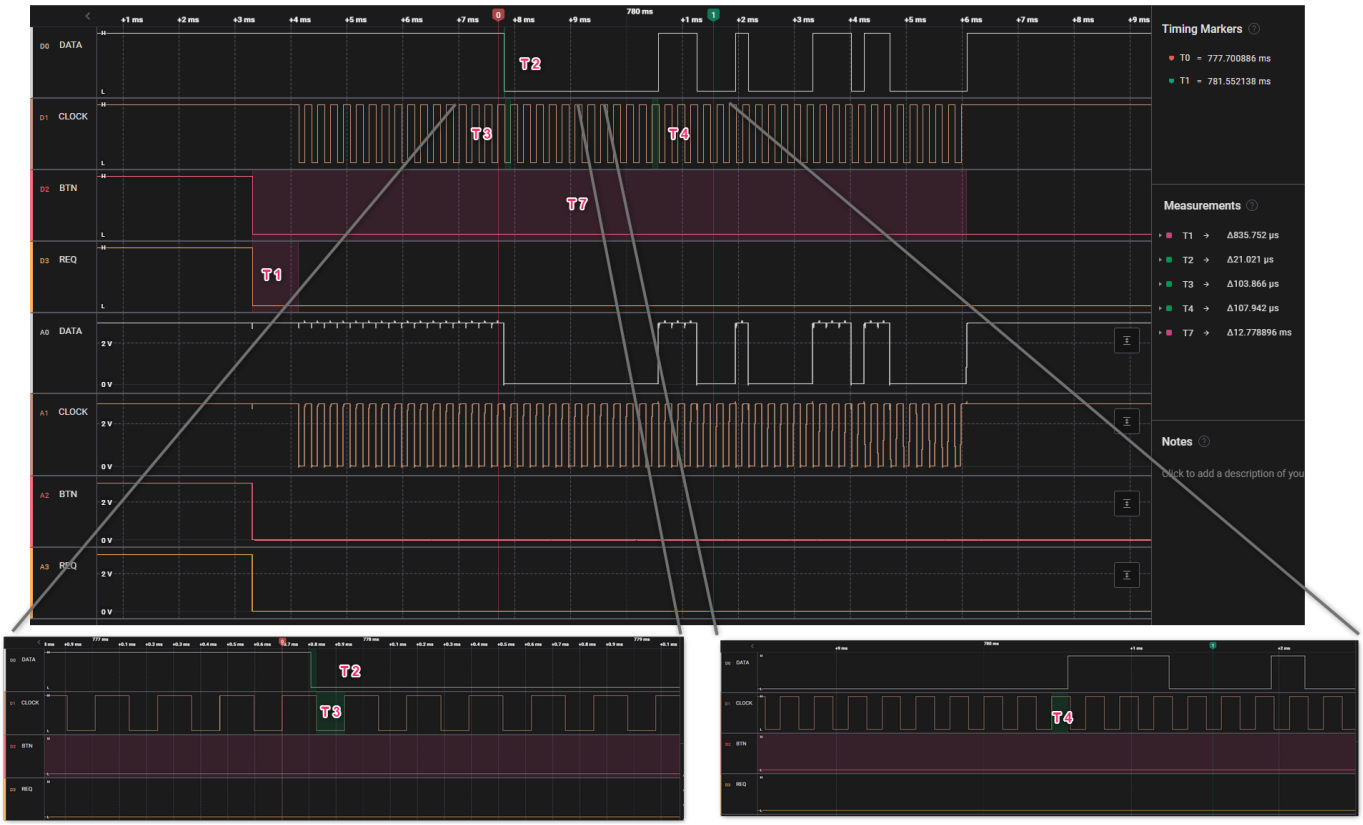




2 ms	≤ T1	≤	40 ms
	T2	=	21 µs (typ.)
	T3	=	100 µs (typ.)
	T4	=	100 µs (typ.)
	T5	=	abhängig von der Leistung des angeschlossenen Datenverarbeitungsgerätes
	T5	=	depends on the performance of the data processing unit
	T6	≤	77 ms
19 ms	≤ T7	≤	57 ms

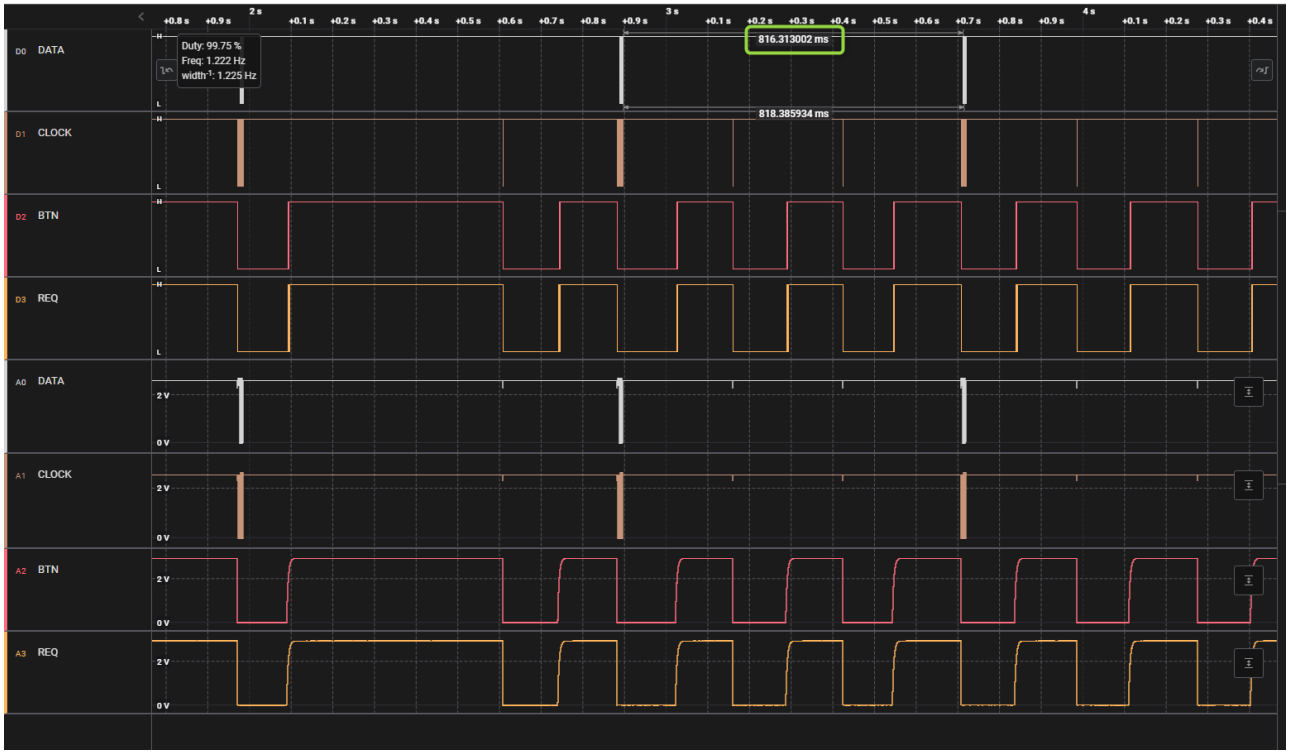
3. Messungen:

3.1. Zeitaufnahme:

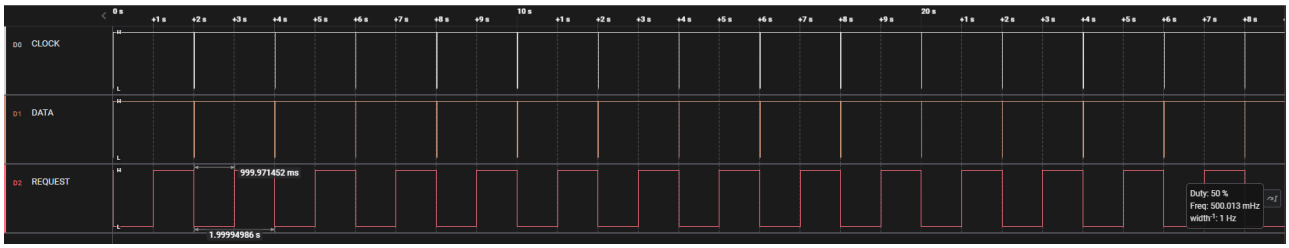


3.2. Zeitaufnahme mit Zycklische-Anforderung:

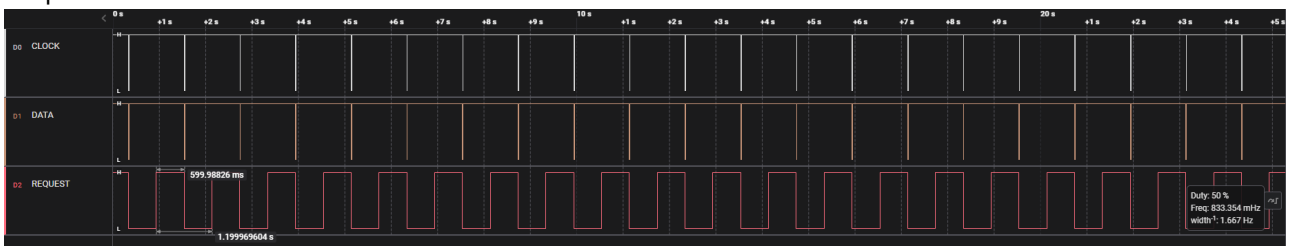
- Digital und Analog:



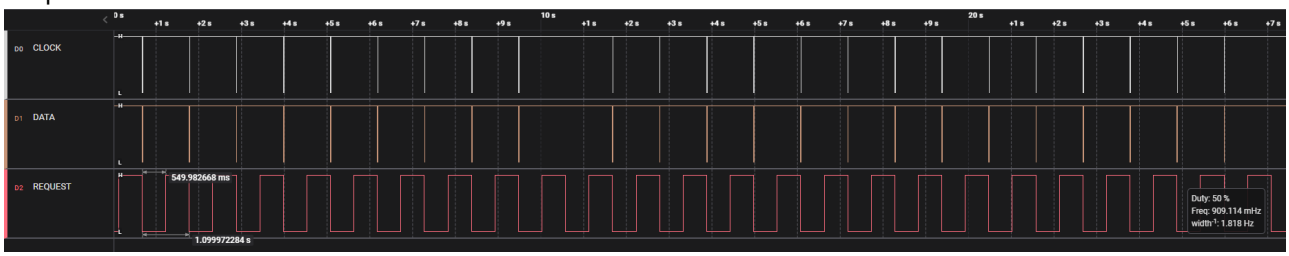
- Requesttakt: 2000ms



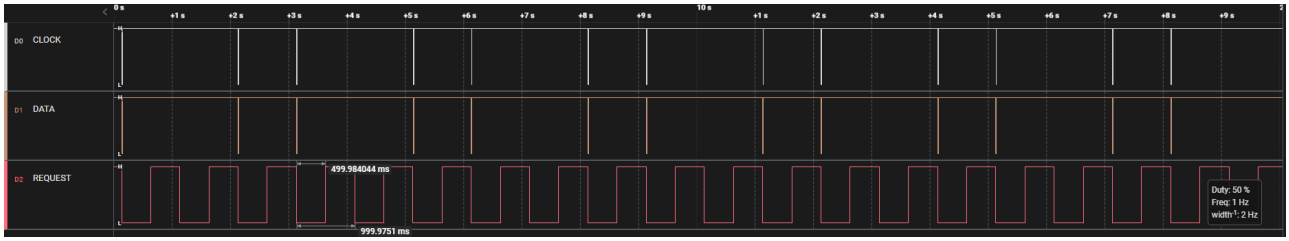
- Requesttakt: 1200ms



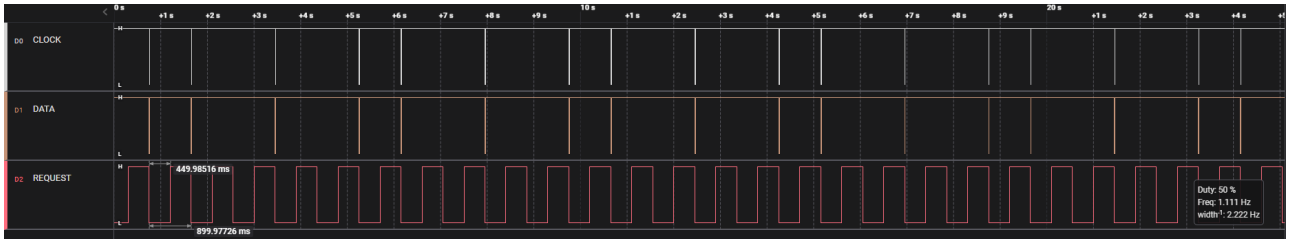
- Requesttakt: 1100ms



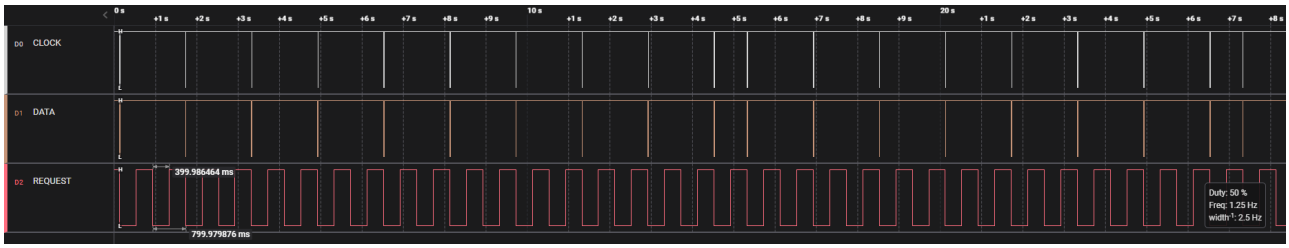
- Requesttakt: 1000ms



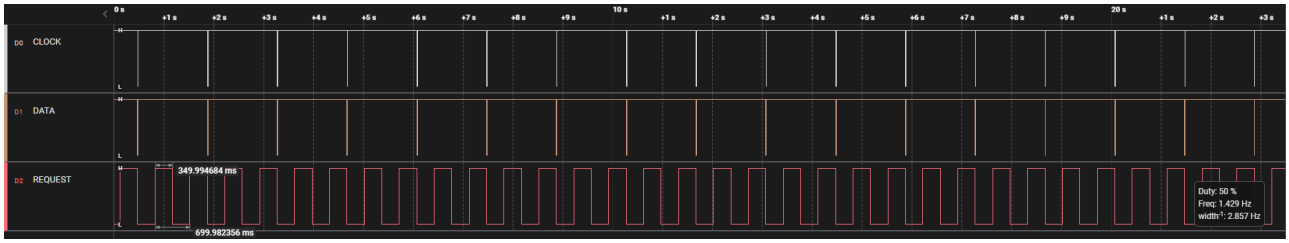
- Requesttakt: 900ms



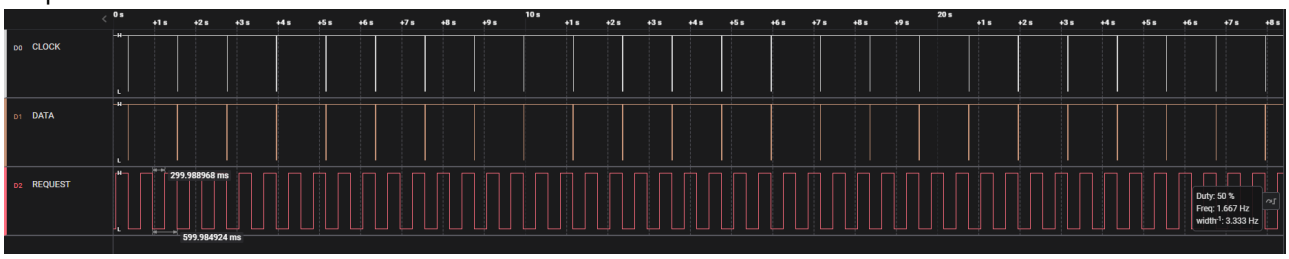
- Requesttakt: 800ms



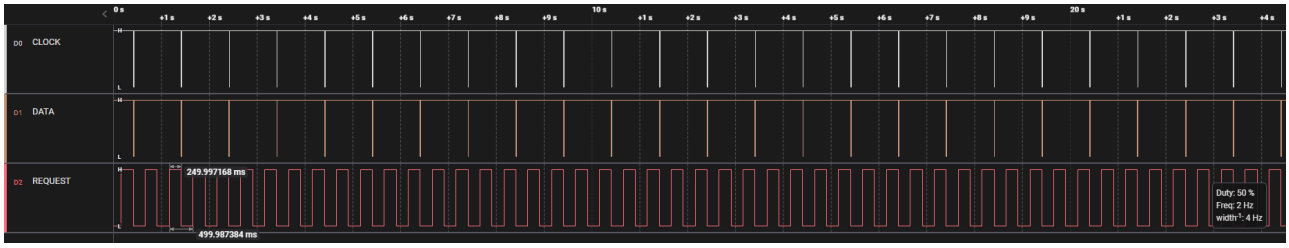
- Requesttakt: 700ms



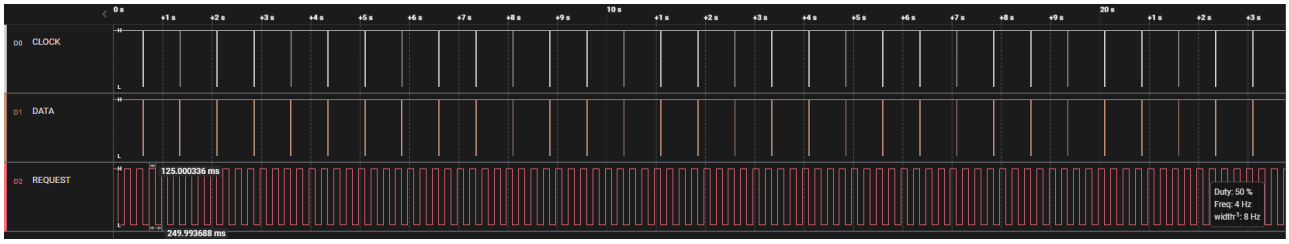
- Requesttakt: 600ms



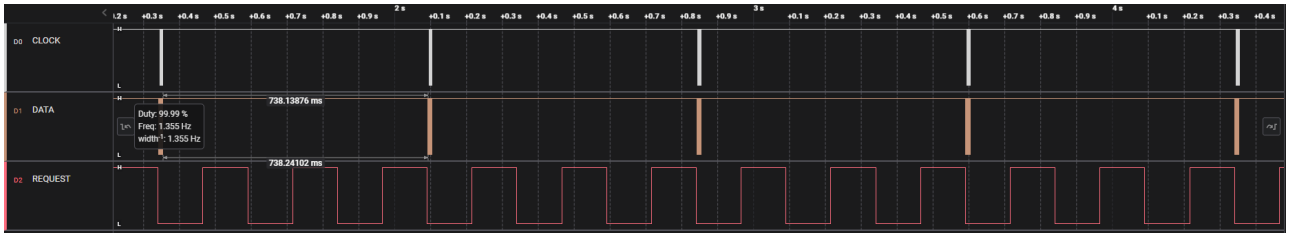
- Requesttakt: 500ms



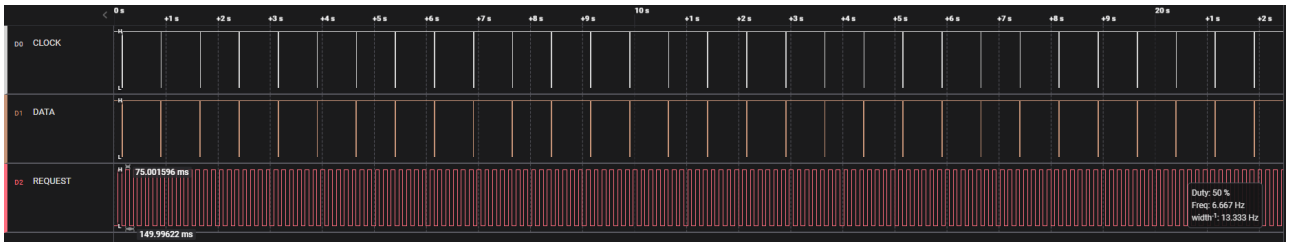
- Requesttakt: 250ms



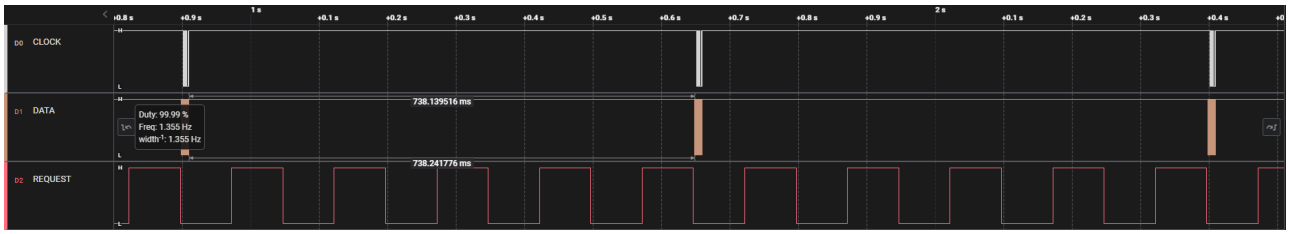
- Requesttakt: 250ms Zoom



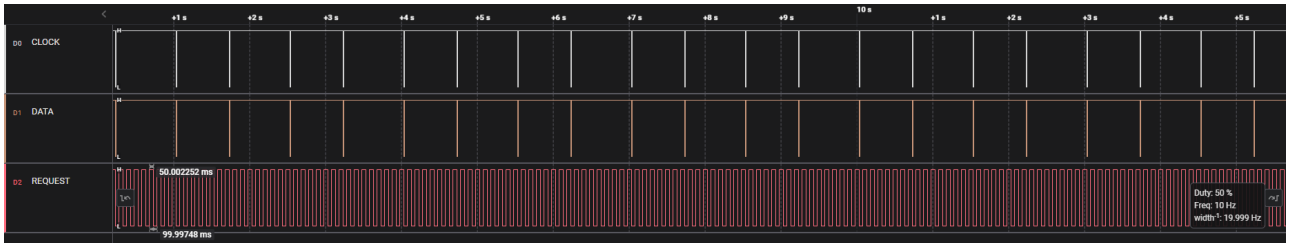
- Requesttakt: 150ms



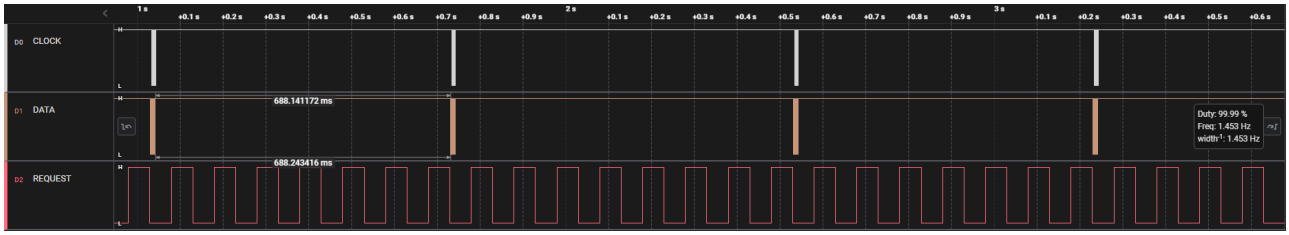
- Requesttakt: 150ms Zoom



- Requesttakt: 100ms



- Requesttakt: 100ms Zoom



4. Ergebnis:

Zeiten T1, T6 und T7 sind auser Toleranz

Zeit	Typ	Min	Max	Ist
T1	-	2 ms	40 ms	0,8 ms
T2	21 us	-	-	21,0 us
T3	100 us	-	-	103,9 us
T4	100 us	-	-	107,9 us
T6	-	-	77 ms	1200 ms
T7	-	19 ms	57 ms	12,8 ms

- 40EWR-L erst bei 1200ms stabil
- 40EWR-L, wenn Anforderungsrate ist schneller (<250ms) antwort stabilisiert sich ca 700ms jede Antwort Sonst datei sind plausiebel. Antwortzeit für Tastendruck ist auch ohne Verzögerung.